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EXAMINATION
MTH301- Calculus II

Question No: 1 (Marks: 1) - Please choose one

π is an example of ----------

► Irrational numbers

► Rational numbers

► Integers

► Natural numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------

► Surface

► Curve

► Plane

► Parabola

Question No: 3 (Marks: 1) - Please choose one

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An ordered triple corresponds to ---------- in three dimensional space.

► A unique point

► A point in each octant

► Three points

► Infinite number of points

Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ----------

► Direction cosines

► Direction ratios

► Direction angles

Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = 4xy + \sin 3x^2 y$

► $f(x, y)$ is continuous at origin

► $f(0, 0)$ is not defined

► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist

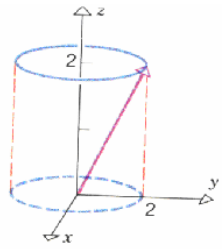
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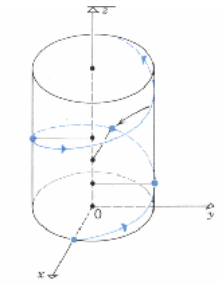
▶ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

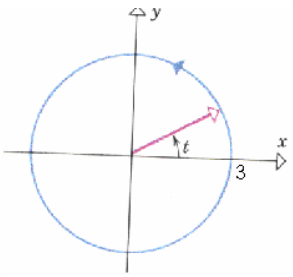
Question No: 6 (Marks: 1) - Please choose one

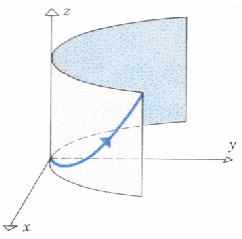
Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ and $0 \le t \le 2\pi$

▶ 

▶ 

▶ 

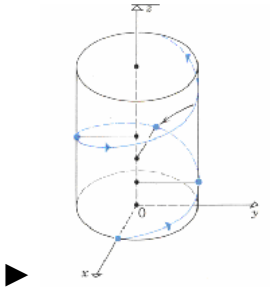
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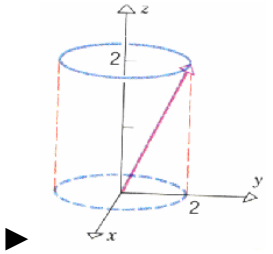
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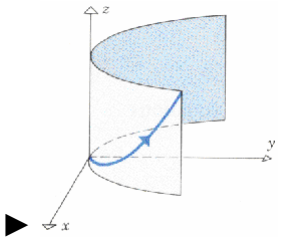
Question No: 7 (Marks: 1) - Please choose one

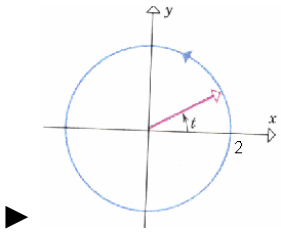
Match the following vector-valued function with its graph.

$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ and $t \geq 0$









Question No: 8 (Marks: 1) - Please choose one

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What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t \hat{i} + (1 - \cos 2t) \hat{j}$

- ▶ $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 0$
- ▶ $y = \sin^2 t \quad , \quad x = 1 - \cos 2t \quad , \quad z = 0$
- ▶ $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 1$
- ▶ $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = 0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

- ▶ $\vec{r}(0)$ is not defined
- ▶ $\vec{r}(0)$ is defined but $\lim_{t \to 0} \vec{r}(t)$ does not exist
- ▶ $\vec{r}(0)$ is defined and $\lim_{t \to 0} \vec{r}(t)$ exists but these two numbers are not equal.
- ▶ $\vec{r}(t)$ is continuous at $t = 0$

Question No: 10 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

- $\vec{r}'(t) = \left(\frac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$
- $\vec{r}'(t) = (\frac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$
- $\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$
- $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2y + 2)\ dx + (x^3 + y)\ dy$

- True
- False

Question No: 12 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 + 4xy)\ dx + (2x^2 + 2y)\ dy$

► True

► False

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

$$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{5}{6} \frac{3}{4} \frac{1}{2}$$

►

$$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$$

►

$$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{6}{7} \frac{4}{5} \frac{2}{3}$$

►

$$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$$

►

Question No: 14 (Marks: 1) - Please choose one

Which of the following is correct?

$$\int_0^{\frac{\pi}{2}} \sin^4 x\, dx = \frac{3}{16}$$

►

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$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$

▶

$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$

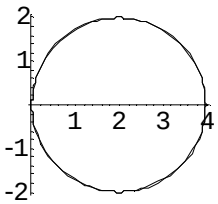
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$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$

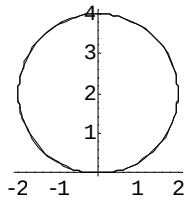
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Question No: 15 (Marks: 1) - Please choose one

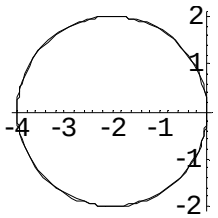
Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$



▶

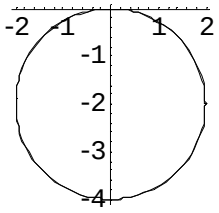


▶



▶

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▶

Question No: 16 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

▶ Initial line

▶ y-axis

▶ Pole

Question No: 17 (Marks: 1) - Please choose one

$$f(x) = \sin\frac{x}{2}$$

What is the period of a periodic function defined by ?

▶ $\frac{\pi}{2}$

▶ π

▶ $\frac{3\pi}{2}$

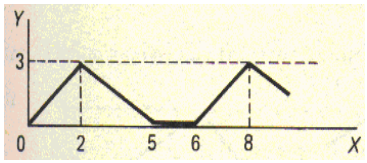
▶ 4π

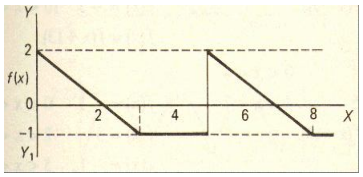
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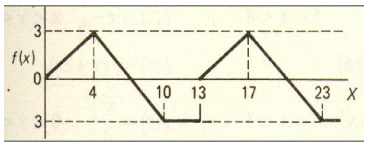
Question No: 18 (Marks: 1) - Please choose one

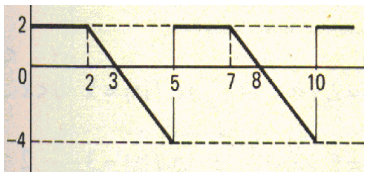
Match the following periodic function with its graph.

$$f(x)=\begin{cases}\frac{3}{4}x & 0<x<4\\ 7-x & 4<x<10\\ -3 & 10<x<13\end{cases}$$

▶ 

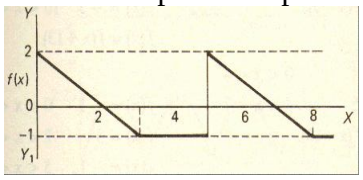
▶ 

▶ 

▶ 

Question No: 19 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

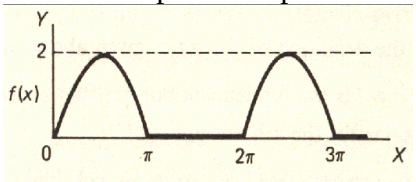
► 3

► 4

► 5

Question No: 20 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 2

► π

► 2π

Question No: 21 (Marks: 1) - Please choose one

$$\left(-2, \frac{-3\pi}{2}\right)$$

Polar co-ordinates of a point are . Which of the following is another possible polar co-ordinates representation of this point?

$$\left(2, \frac{-\pi}{4}\right)$$

►

- $\left(2, \frac{-\pi}{2}\right)$
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is ----------

- Even function
- Odd function
- Neither even nor odd

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ----------

- x-axis
- y-axis
- origin

Question No: 24 (Marks: 1) - Please choose one

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At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

- (0, 3)
- (2, −1)
- (−2, 15)
- (1, 0)

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

- (2, − 2)
- (−4, 34)
- (0, 0)
- (−2, 14)

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

- $f(x, y)$ is continuous at origin
- $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist

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- $f(0, 0)$ is defined and $\lim_{(x, y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when ----------

- path of integration is divided into parts.
- path of integration is parallel to y-axis.
- direction of path of integration is reversed.
- path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

- $\int_0^s e^{-st} F(t)\, dt$
- $\int_0^\infty e^{st} F(t)\, dt$
- $\int_{-\infty}^\infty e^{-st} F(t)\, dt$
- $\int_0^\infty e^{-st} F(t)\, dt$

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Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

- ▶ $L\{e^{5t}\} = \frac{1}{s-5}$
- ▶ $L\{e^{5t}\} = \frac{s}{s^2+25}$
- ▶ $L\{e^{5t}\} = \frac{5}{s^2+25}$
- ▶ $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\frac{1}{s+1}$

- ▶ $L^{-1}\left\{\frac{1}{s+1}\right\} = t+1$
- ▶ $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}+e^{t}$
- ▶ $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{t}$
- ▶ $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

- ▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$
- ▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$
- ▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$
- ▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

Question No: 32 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

- ▶ $L\{-6\} = \frac{1}{s+6}$
- ▶ $L\{-6\} = \frac{-6}{s}$
- ▶ $L\{-6\} = \frac{s}{s^2+36}$
- ▶ $L\{-6\} = \frac{-6}{s^2+36}$

Question No: 33 (Marks: 1) - Please choose one

$$\int_C (3x+2y)\,dx+(2x-y)\,dy$$

Evaluate the line integral where C is the line segment from (0, 0) to (2, 0).

► 6

► -6

► 0

► Do not exist

Question No: 34 (Marks: 1) - Please choose one

$$\int_C (2x+y)\,dx+(x^2-y)\,dy$$

Evaluate the line integral where C is the line segment from (0, 0) to (0, 2).

► -4

► -2

► 0

► 2

Question No: 35 (Marks: 1) - Please choose one

Plane is an example of ----------

► Curve

► Surface

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- Sphere
- Cone

Question No: 36 (Marks: 1) - Please choose one

$If\ R = \{(x, y) / 0 \le x \le 2\, and\, -1 \le y \le 1\}, then$

$\iint_R (x + 2y^2) dA =$

- $\int_{-1}^{1} \int_{0}^{2} (x + 2y^2) dy dx$
- $\int_{0}^{2} \int_{1}^{-1} (x + 2y^2) dx dy$
- $\int_{-1}^{1} \int_{0}^{2} (x + 2y^2) dx dy$
- $\int_{1}^{2} \int_{-1}^{0} (x + 2y^2) dx dy$

Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

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- $dr = dx\,\hat{i} + dy\,\hat{j}$

- $dr = dx\,\hat{i} + dy\,\hat{j} + dy\,\hat{k}$

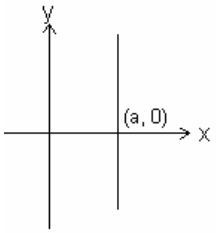
- $dr = dx + dy + dz$

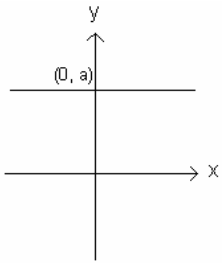
- $dr = dx + dy$

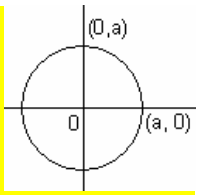
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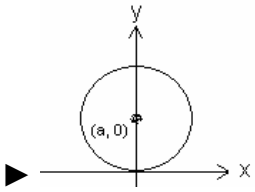
Match the following equation in polar co-ordinates with its graph.
$r = a$

where a is an arbitrary constant.

- 

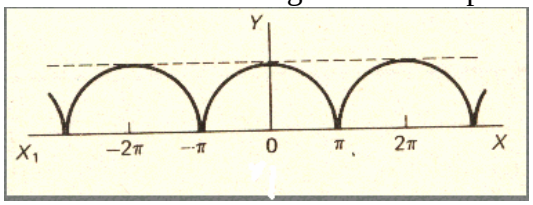
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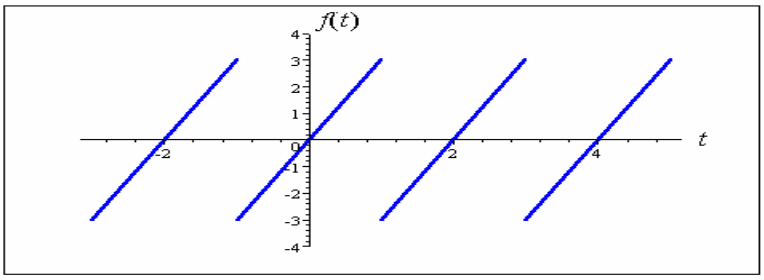
Question No: 39 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?



- Even function
- Odd function
- Neither even nor odd function

Question No: 40 (Marks: 1) - Please choose one



The graph of “saw tooth wave” given above is ----------

- An odd function

- An even function
- Neither even nor odd

Question No: 1 (Marks: 2) - Please choose one

Laplace transform of 't ' is

- $\frac{1}{s}$
- $\frac{1}{s^2}$
- e^{-s}
- s

Question No: 2 (Marks: 2) - Please choose one

Symmetric equation for the line through (1,3,5) and (2,-2,3) is

- $x-2=-\frac{y+2}{3}=-\frac{z-3}{5}$
- $x+2=-\frac{y+3}{5}=-\frac{z+5}{2}$
- $x-1=-\frac{y-3}{5}=-\frac{z-5}{2}$
- $x+1=\frac{y+3}{5}=\frac{z-5}{5}$

Question No: 3 (Marks: 1) - Please choose one

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The level curves of f(x, y) = y Cscx are parabolas.

- True.
- False.

Question No: 4 (Marks: 1) - Please choose one

The equation $z = r$ is written in

- Rectangular coordinates
- Cylindrical coordinates
- Spherical coordinates
- None of the above